complicated techniques. The energy of the information signal is sufficient enough to be sent directly. However if the information in the audio frequency range (20 to 20,000 Hz) needs to be transmitted to long distances across the world, certain techniques are required to transmit the information without any loss.

For long distance transmission, the low frequency baseband signal (input signal) is superimposed onto a high frequency radio signal by a process called modulation. In the modulation process, a very high frequency signal called carrier signal (radio signal) is used to carry the baseband signal.

As the frequency of the carrier signal is very high, it can be transmitted to long distances with less attenuation. The carrier signal is usually a sine wave signal.

A sinusoidal carrier wave can be represented as $e_c = E_c \sin(2\pi v_c t + \phi)$, where E_c is the amplitude, v_c is the frequency and ϕ is the initial phase of the carrier wave at any instant of time t.

Three characteristics in the carrier signal can be modified by the baseband signal during the process of modulation: amplitude, frequency and phase of the carrier signal.

Accordingly, we have (i) amplitude modulation, (ii) frequency modulation and (iii) phase modulation.

10.9.1 AMPLITUDE MODULATION (AM)

If the amplitude of the carrier signal is modified in proportion to the instantaneous amplitude of the baseband signal, then it is called amplitude modulation. Here the frequency and the phase of the carrier signal remain constant. Amplitude modulation is used in radio and TV broadcasting.

The signal shown in Figure 10.50(a) is the baseband signal that carries information. Figure 10.50(b) shows the high-frequency carrier signal and Figure 10.50(c) gives amplitude modulated signal. We can see that amplitude of the carrier wave is modified in proportion to the amplitude of the baseband signal.

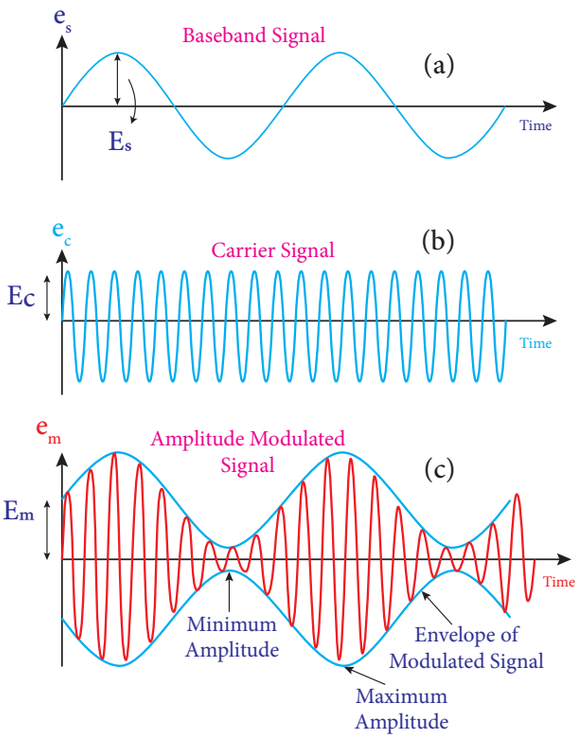


Figure 10.50 Amplitude Modulation (a) Baseband signal (b) Carrier signal (c) Modulated signal

Advantages of AM

i) Easy transmission and reception
ii) Lesser bandwidth requirements
iii) Low cost

Limitations of AM

i) Noise level is high
ii) Low efficiency
iii) Small operating range

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